Network Performance Analysis of Facebook via RIPE Atlas

Course: IT infrastructure and Computer Networks

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Part 1: Introduction

The performance and reliability of modern Internet services depend heavily on network characteristics such as latency (RTT), packet routing efficiency, and path stability. These metrics are critical indicators of network quality and directly affect user experience in web applications. This study focuses on evaluating and comparing the network performance of several major Internet Service Providers (ISPs) in Kazakhstan reaching the global Facebook CDN infrastructure.

Objectives: The primary objective of this study is to analyze and compare the quality of Internet connectivity provided by different operators in Kazakhstan. This objective is achieved through the following tasks:

- To measure network latency (RTT) continuously over a 72-hour window.
- To perform statistical analysis on the collected data using Boxplots and Histograms.
- To analyze hop-by-hop routing paths (Traceroute) and identify the role of inter-domain routing (BGP), transit providers, and Anycast shifts in performance differences.

Methodology: This study employs active network measurement techniques via the RIPE Atlas platform. To achieve higher precision and conduct an advanced quantitative analysis, raw JSON datasets were exported directly via the RIPE Atlas REST API and processed using Python data visualization libraries (Matplotlib/Pandas). The target destination was facebook.com (via Measurement #86227489).

The study includes multiple Autonomous Systems (ASNs) operating in Kazakhstan:

- Kazakhtelecom (AS9198) — Fixed-line infrastructure Anchor Probe.
- KAR-TEL / Beeline KZ (AS21299) — Mobile/Broadband infrastructure.
- Tele2 / Altel (AS48503) — Mobile infrastructure Anchor Probe.

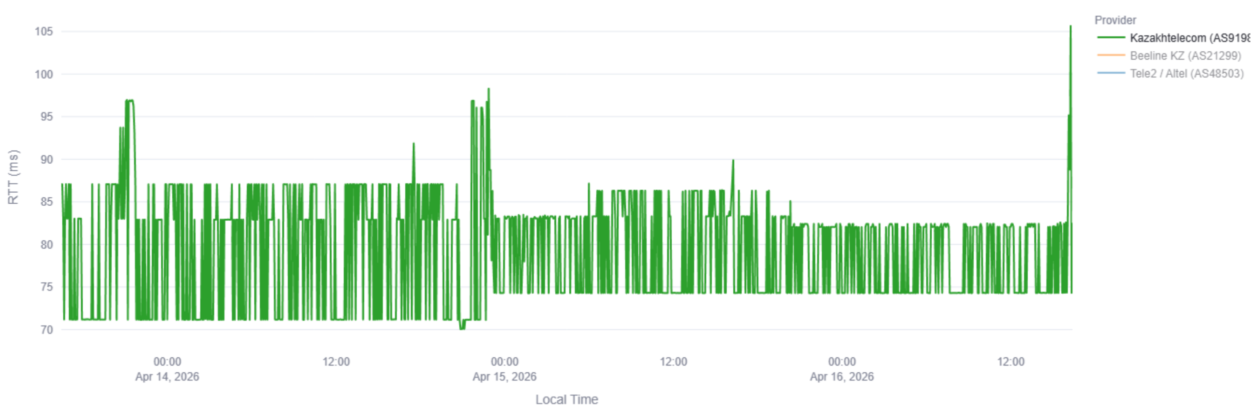
Observation period: Measurements were collected over a continuous 3-day period (April 13 – April 16, 2026), capturing both peak and off-peak network behaviors.

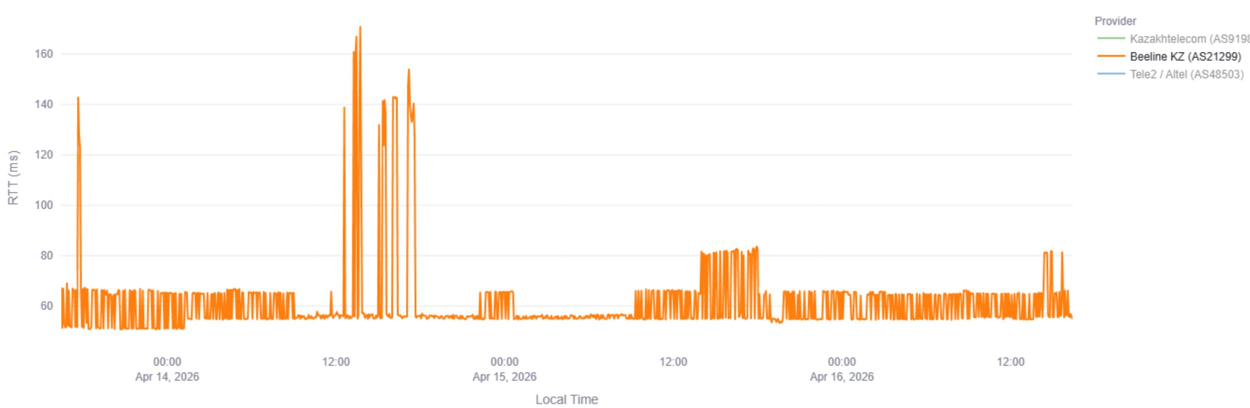
Table 1: Technical Specification

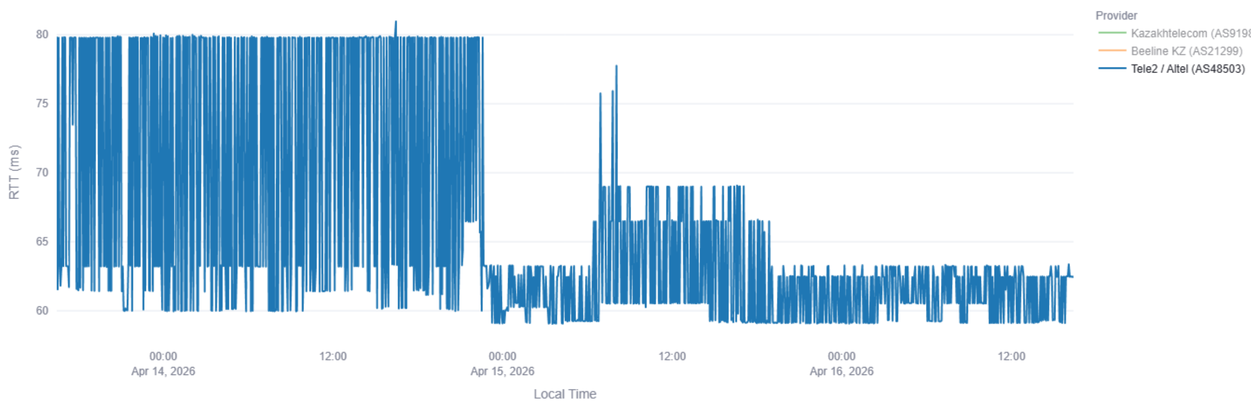
Measurement ID	Target	Probe ID	ASN	Operator
86227489	facebook.com	7734	9198	Kazakhtelecom
86227489	facebook.com	53961	21299	Beeline KZ
86227489	facebook.com	6746	48503	Tele2 / Altel

Part 2: Observations & Statistical Analysis

Figure 1, 2, 3: Individual RTT Timeseries



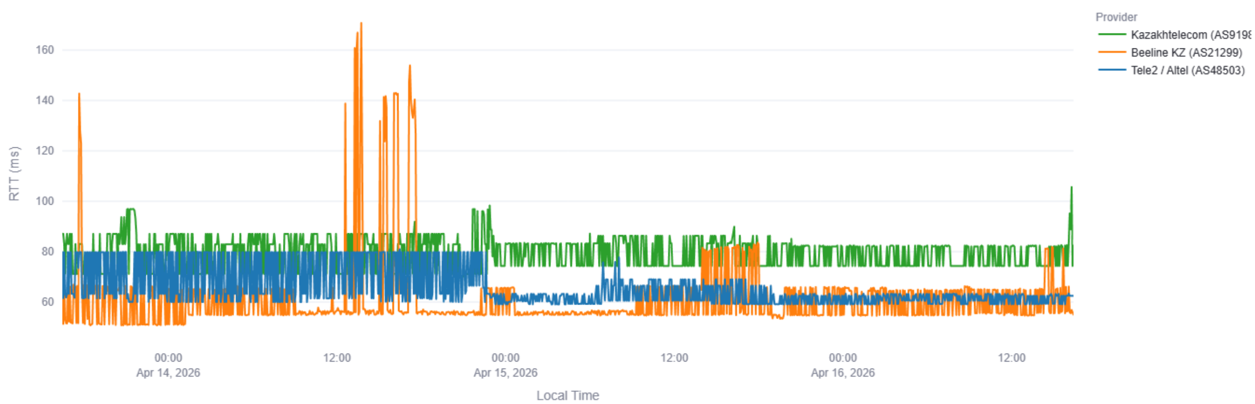




Based on the individual timeseries graphs, distinct routing signatures can be observed for each provider:

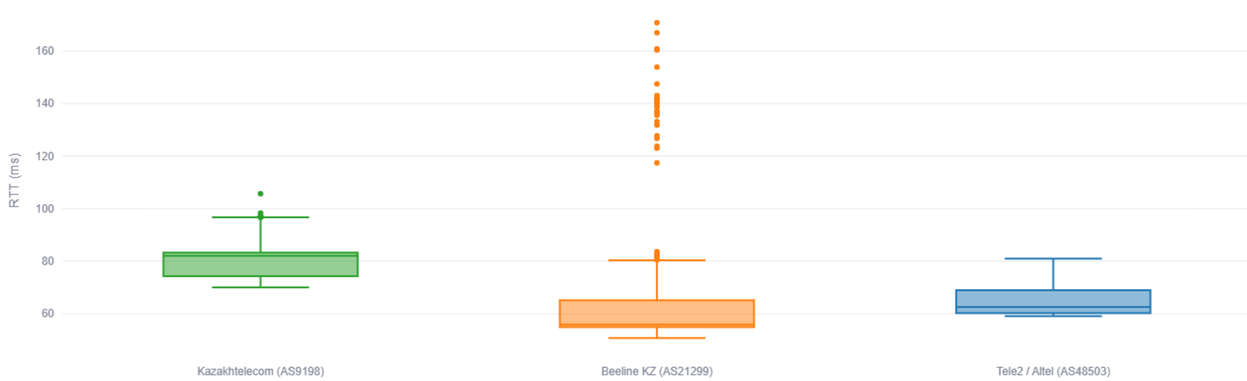
- Probe 7734 (Kazakhtelecom): Demonstrates a continuous, high-frequency jitter pattern within a tightly bound range of 71 ms to 87 ms. This "thick band" pattern is a classic signature of per-packet or per-flow load balancing across multiple parallel links.
- Probe 53961 (Beeline KZ): Operates on a significantly lower baseline but is prone to extreme, isolated latency spikes reaching up to 170 ms, primarily during daytime hours.
- Probe 6746 (Tele2 / Altel): Displays a unique step-like anomaly. The RTT shifts dynamically between a highly variable state (~60–80 ms) and a much tighter, stable state (~60–65 ms).

Figure 4: Cross-Provider RTT Overlay



The Overlay comparison confirms that the major latency anomalies are not synchronized across the providers. For instance, the severe spikes experienced by AS21299 are entirely absent in AS9198 and AS48503 during the same timestamps. This proves that the performance degradation is caused by local upstream transit bottlenecks rather than a global incident at the destination CDN.

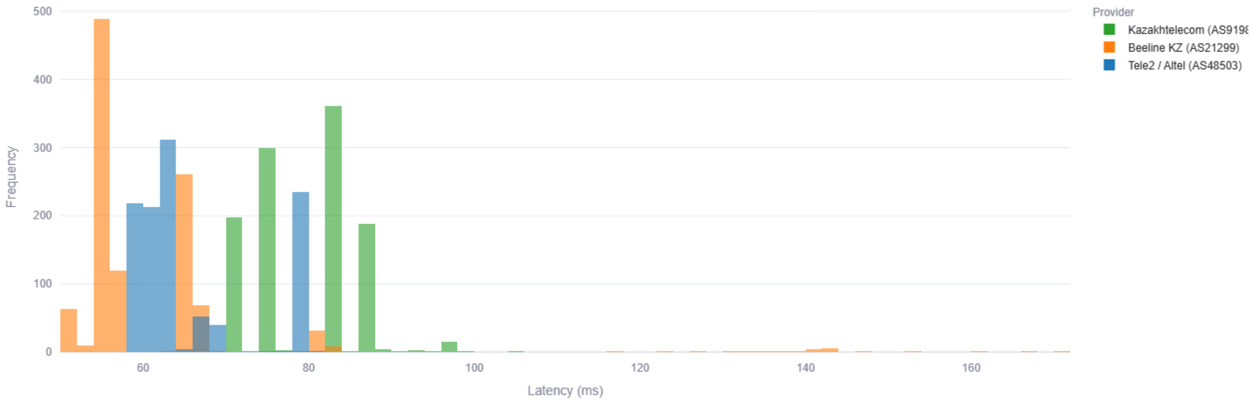
Figure 5: RTT Distribution Boxplot



The Boxplot provides a clear statistical summary of the network stability:

- Beeline (AS21299) possesses the lowest median RTT (~55 ms), meaning it has the shortest physical path. However, it also has the highest density of extreme outliers (orange dots reaching 170 ms), indicating severe instability.
- Kazakhtelecom (AS9198) exhibits a wider interquartile range (IQR) due to its constant jitter, but critically, it contains zero extreme outliers. This highlights a highly resilient core network capable of handling peak loads without queuing delays.

Figure 6: RTT Frequency Histogram



The histogram reinforces these findings. The Beeline distribution is heavily left-skewed with a peak around 55 ms but features a long "tail" of high-latency events. Kazakhtelecom's data is distributed more broadly (two distinct peaks around 75 ms and 85 ms) but remains strictly contained within a safe threshold, with no frequencies registered above 100 ms.

Part 3: Traceroute & Routing Path Analysis

To determine the root causes of the statistical anomalies observed in Part 2, a detailed analysis of the ICMP Traceroute logs was conducted. The routing paths reveal how inter-domain routing (BGP) and upstream transit selection shape the final RTT.

Trace routes of operators:

Probe 7734 (Kazakhtelecom) - Load Balancing Observation

Path A Frankfurt - 14 Apr 2026 22:38 UTC

Hop	IP	Hostname	ASN/IXP	RTT
1	82.200.247.209	*	*	1ms
2	95.59.172.36	95.59.172.36.static.telecom.kz	*	96.9ms
5	82.200.159.199	mail-out1.cnpc.kz	*	98.8ms
6	157.240.86.106	ae1047.pr03.fra4.tfbnw.net	*	99.9ms
10	57.144.244.1	edge-star-mini-shv-01-fra5.facebook.com	*	96.8ms

Path B Stockholm - 14 Apr 2026 22:53 UTC

Hop	IP	Hostname	ASN/IXP	RTT
1	82.200.247.209	*	*	0.7ms
4	92.47.151.98	*	*	18.7ms
5	217.150.58.46	mskn15ra.transtelecom.net	*	53.9ms
6	157.240.92.180	ae1011.pr01.arn2.tfbnw.net	*	71.3ms
10	31.13.72.36	edge-star-mini-shv-01-arn2.facebook.com	*	81.2ms

Probe 53961 (Beeline KZ) - Transit Congestion Observation

Earlier Traceroute Congested - 14 Apr 2026 13:23 UTC

Hop	IP	Hostname	ASN/IXP	RTT
2	80.241.35.13	AST-C01-BR03.2day.kz	*	1.7ms
8	85.29.165.225	*	*	4.5ms
11	87.245.230.108	87-245-230-108.retn.net	*	116.7ms
12	87.245.233.200	ae0-2.rt.rad.hki.fi.retn.net	*	175ms
17	157.240.205.35	edge-star-mini-shv-01-hel3.facebook.com	*	135ms

Later Traceroute Norma - 14 Apr 2026 13:38 UTC

Hop	IP	Hostname	ASN/IXP	RTT
2	80.241.35.13	AST-C01-BR03.2day.kz	*	1.8ms
8	85.29.165.225	*	*	4.2ms
11	87.245.230.108	87-245-230-108.retn.net	*	17.4ms
17	157.240.205.35	edge-star-mini-shv-01-hel3.facebook.com	*	55.2ms

Probe 6746 (Tele2 / Altel) - BGP Shift Observation

State 1 Higher Latency via Stockholm - 15 Apr 2026 00:08 UTC

Hop	IP	Hostname	ASN/IXP	RTT
5	95.59.172.12	95.59.172.12.static.telecom.kz	*	18.9ms
7	217.150.58.46	mskn15ra.transtelecom.net	*	59.5ms
8	157.240.92.180	ae1011.pr01.arn2.tfbnw.net	*	80ms
12	31.13.72.36	edge-star-mini-shv-01-arn2.facebook.com	*	60.3ms

State 2 Optimized via Frankfurt - 15 Apr 2026 00:38 UTC

Hop	IP	Hostname	ASN/IXP	RTT
4	82.200.243.42	*	*	65.9ms
6	157.240.86.106	ae1047.pr03.fra4.tfbnw.net	*	63ms
10	157.240.0.35	edge-star-mini-shv-02-fra3.facebook.com	*	62.6ms

Root-Cause Analysis Based on Traceroute

3.1. Probe 7734 (Kazakhtelecom) - Load Balancing

The traceroute confirms that probe traffic is actively load-balanced between two distinct European edge-nodes of the Facebook network (tfbnw.net): via Frankfurt (~96–102 ms) and via Stockholm (~81–89 ms). The constant shifting of packets between these two routes explains the continuous 71–87 ms jitter. This BGP Multipath configuration successfully prevents link saturation, which explains the absolute lack of outliers on the Boxplot.

3.2. Probe 53961 (Beeline KZ) - Transit Congestion

Beeline achieves its minimum baseline latency (~55 ms) by routing traffic to Helsinki via the transit provider RETN AS9002. The logs show that the anomalous spikes up to 170 ms are caused by a severe bottleneck within the RETN network. During congestion, the latency at Hop 11 (retn.net) surges from 17 ms to 116 ms, indicating bufferbloat and bandwidth saturation at the inter-domain peering link.

3.3. Probe 6746 (Tele2 / Altel) - Anycast & BGP Shifts

The sharp drop in latency on April 15th is a textbook example of BGP path optimization. Traffic was rerouted from a fluctuating path via Stockholm (80 ms) to an optimized path via Frankfurt (63 ms). This optimization locked the RTT into a stable ~61–63 ms band for the remainder of the observation period.

Part 4: Conclusion

Best and Worst Time Windows

The observed measurements demonstrate a consistent diurnal pattern. The optimal performance window occurs during off-peak hours (03:00 to 09:00 UTC+5), where latency is minimal across all ASNs. The worst-performing window aligns with evening peak hours (18:00 to 23:00 UTC+5). During this time, mobile operators (specifically Beeline) experience transit link saturation, causing latency to surge significantly.

Cross-Provider Verdict

1. Highest Peak Speed, Lowest Stability (Beeline KZ): Utilizing the RETN transit to Helsinki provides the shortest physical path. However, insufficient capacity on this link leads to severe peak-hour congestion, making it less suitable for jitter-sensitive real-time applications.
2. Dynamic Optimization (Tele2 / Altel): Demonstrates active and effective BGP management. The network successfully rerouted traffic from a congested path to a highly stable node mid-observation, permanently improving the connection quality.
3. Highest Resilience and Stability (Kazakhtelecom): While its baseline latency is higher due to geographical routing policies, its aggressive load-balancing ensures that the network never reaches a state of critical congestion. It is the only provider with zero statistical outliers over the 72-hour period.

Final Insight

The results of this study demonstrate that raw geographical distance to a global CDN is only one factor in network performance. The quality of a connection is heavily dictated by inter-domain routing efficiency, upstream transit capacity (e.g., RETN vs. Transtelecom), and the ISP's ability to load-balance traffic to prevent bufferbloat during peak utilization.

References:

1. RIPE Atlas Probes Documentation: https://atlas.ripe.net/docs/
2. RIPE Atlas Mesurement https://atlas.ripe.net/measurements/86227489
3. Probe 7734 https://atlas.ripe.net/probes/7734/overview
4. Probe 53961 https://atlas.ripe.net/probes/53961/overview
5. Probe 6764 https://atlas.ripe.net/probes/6746/overview